

HOUSE PRICE PREDICTION USING LINEAR REGRESSION **ON CALIFORNIA HOUSING DATASET**

Introduction

Machine Learning is widely used to predict future outcomes from historical data. Regression algorithms are commonly used when the target variable is continuous. In this project, a Linear Regression model was developed to predict house prices using the California Housing Dataset.

The main objective of this project was to understand the machine learning workflow, including data loading, exploratory data analysis (EDA), preprocessing, model training, prediction, and evaluation.

Machine learning

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that enables computers to learn patterns from data and make predictions or decisions without being explicitly programmed for every task.

In machine learning, algorithms improve their performance automatically as they are exposed to more data.

Example: A house price prediction model learns the relationship between house features and prices from historical data and then predicts prices for new houses.

Types of Machine Learning:

1. Supervised Learning

In supervised learning, the model is trained using labeled data, where both input and correct output are provided.

Applications:

- House price prediction
- Email spam detection
- Weather forecasting

Algorithms:

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forest

Example:

Predicting house prices using features such as income, house age, and location.

2. Unsupervised Learning

In unsupervised learning, the model is trained using unlabeled data and tries to discover hidden patterns or groups.

Applications:

- Customer segmentation
- Market basket analysis
- Data clustering

Algorithms:

- K-Means Clustering
- Hierarchical Clustering
- Principal Component Analysis (PCA)

Example:

Grouping customers with similar purchasing behavior.

3. Reinforcement Learning

In reinforcement learning, an agent learns by interacting with an environment and receiving rewards or penalties for its actions.

Applications:

- Robotics
- Self-driving cars
- Game playing

Example:

A robot learns the best path to reach a destination by receiving rewards for correct movements.

Linear Regression

Linear Regression is a supervised machine learning algorithm used to predict a continuous numerical value by establishing a linear relationship between input features and the target variable. It works by fitting a straight line, known as the **line of best fit**, through the data points such that the overall prediction error is minimized.

The line of best fit represents the relationship between the independent variables and the dependent variable and is mathematically expressed as:

$$\text{Predicted Value} = \text{Intercept} + (\text{Coefficient} \times \text{Feature}_1) + (\text{Coefficient} \times \text{Feature}_2) + \dots$$

The model determines the optimal coefficients and intercept that minimize the difference between the actual and predicted values.

Linear Regression was selected for this assignment because:

1. It is one of the simplest and most widely used regression algorithms.
2. The target variable, house price, is a continuous numerical value, making it suitable for regression analysis.
3. It is easy to implement and interpret.
4. It provides a strong baseline model for evaluating prediction performance.
5. It allows performance assessment using standard regression metrics such as MAE, RMSE, and R² Score.

Therefore, Linear Regression was used to learn the relationship between housing features such as median income, house age, location, and population, and predict the median house value for new observations.

Dataset Description

The California Housing Dataset was obtained from the Scikit-learn library. The dataset contains information collected from California districts.

Features

- MedInc (Median Income)
- HouseAge (Median House Age)
- AveRooms (Average Rooms)
- AveBedrms (Average Bedrooms)
- Population
- AveOccup (Average Occupancy)

- Latitude
- Longitude

Target Variable

- MedHouseVal (Median House Value)

The dataset was loaded into a Pandas DataFrame for further analysis.

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude	MedHouseVal
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	-122.23	4.526
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	-122.22	3.585
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	-122.24	3.521
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85	-122.25	3.413
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85	-122.25	3.422
...	...	...	...	...	...	...	...	...	...
20635	1.5603	25.0	5.045455	1.133333	845.0	2.560606	39.48	-121.09	0.781
20636	2.5568	18.0	6.114035	1.315789	356.0	3.122807	39.49	-121.21	0.771
20637	1.7000	17.0	5.205543	1.120092	1007.0	2.325635	39.43	-121.22	0.923
20638	1.8672	18.0	5.329513	1.171920	741.0	2.123209	39.43	-121.32	0.847
20639	2.3886	16.0	5.254717	1.162264	1387.0	2.616981	39.37	-121.24	0.894

20640 rows × 9 columns

Exploratory Data Analysis (EDA) of California Housing Dataset

Exploratory Data Analysis (EDA) is an important step in the machine learning workflow. It involves examining the dataset using statistical summaries and visualizations to understand its structure, identify patterns, detect anomalies, and determine relationships among variables.

The California Housing Dataset contains information about housing districts in California. The target variable is MedHouseVal (Median House Value), while the input features include median income, house age, average rooms, population, occupancy, latitude, and longitude.

The dataset was first inspected using:

- `df.head()` to view the first five records.

- `df.info()` to examine data types and structure.
- `df.describe()` to obtain summary statistics.
- `df.isnull().sum()` to check for missing values.

Inference:

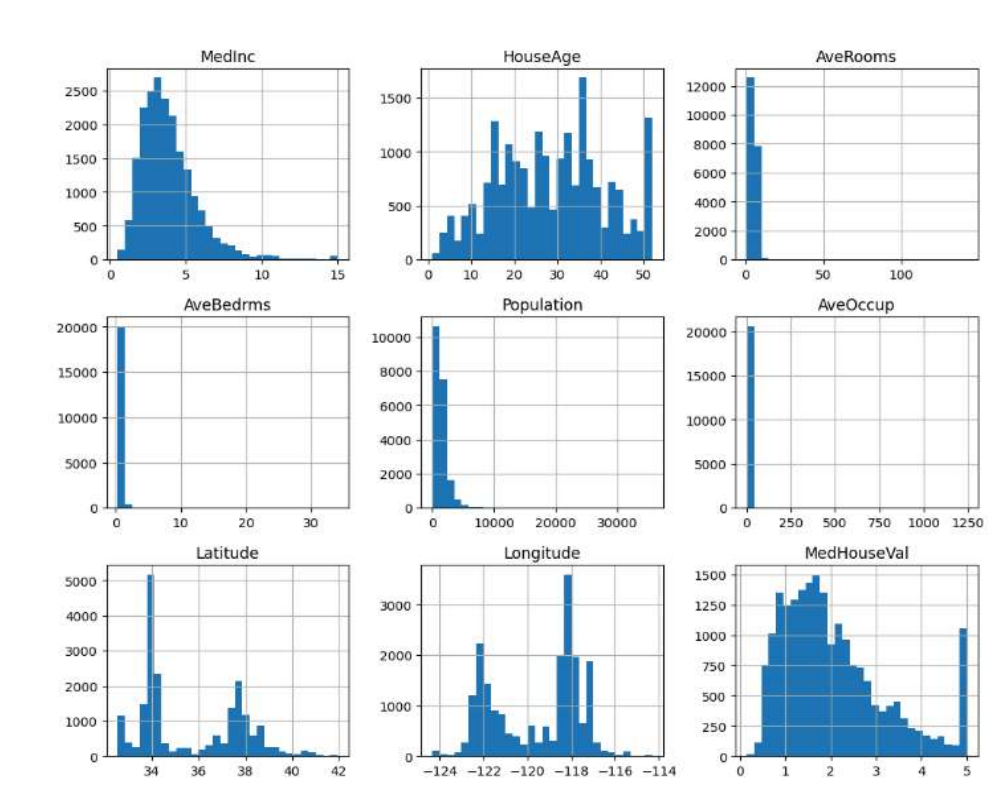
The dataset contains only numerical features and is suitable for regression analysis. No significant missing values were observed, indicating that the dataset is clean and ready for model training.

Feature Distribution Histograms

Histograms were generated using:

```
df.hist(figsize=(12,10), bins=30)
```

The purpose of the histogram is to visualize the distribution of each feature and identify skewness, concentration of values, and potential outliers.



Observations:

- Median Income (MedInc) showed a right-skewed distribution, indicating that most districts have moderate incomes while a few have very high incomes.
- House Age values were concentrated within a specific range.
- Population and Average Occupancy displayed significant variation across districts.
- Some features exhibited extreme values, suggesting the presence of outliers.

Inference:

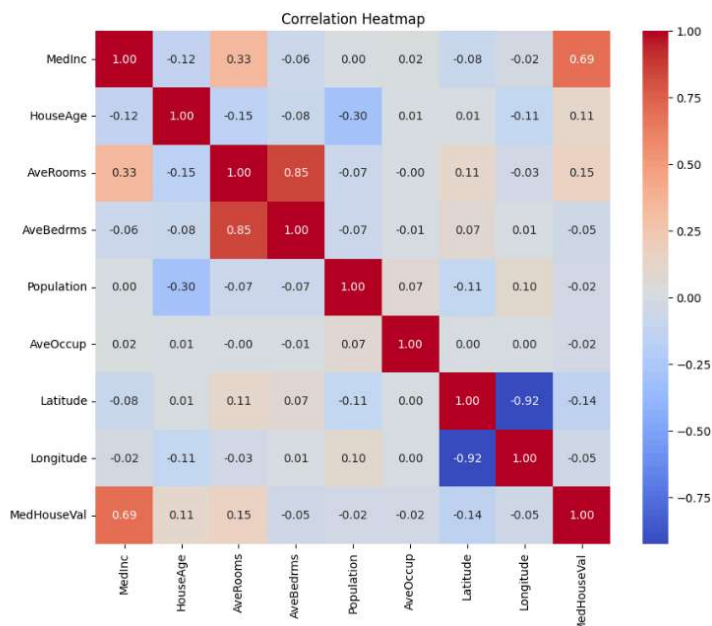
The histogram analysis helped understand the spread and frequency of each variable. It also indicated that certain variables may require additional preprocessing if more advanced models are used in future work.

Correlation Heatmap

A correlation heatmap was created using:

```
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
```

The heatmap displays correlation coefficients between pairs of variables and helps identify relationships among features.



Observations:

- Median Income (MedInc) exhibited the strongest positive correlation with Median House Value (MedHouseVal).
- Latitude and Longitude showed noticeable correlations with house prices, indicating geographical influence.
- Some features displayed weak correlations with the target variable.

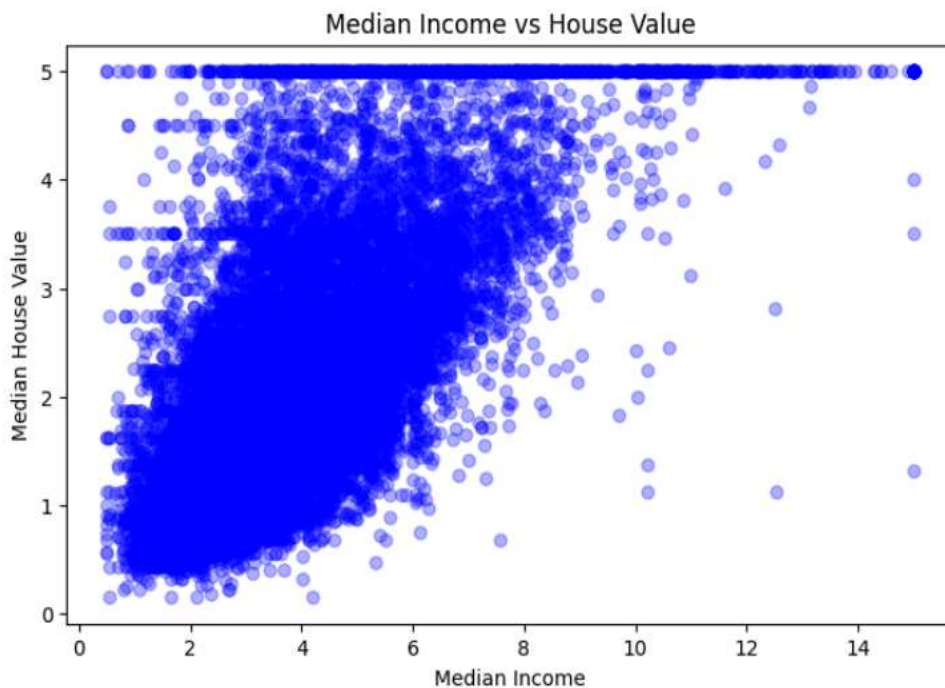
Inference:

The heatmap revealed that income is the most influential factor affecting house prices. This insight supports the expectation that areas with higher incomes generally have higher property values. Understanding these relationships is useful when selecting features for model training.

Scatter Plot: Median Income vs House Value

A scatter plot was generated to visualize the relationship between median income and house value.

```
plt.scatter(df['MedInc'], df['MedHouseVal'])
```



Observations:

- A clear upward trend was observed.
- Districts with higher median incomes generally had higher house prices.
- Some variation existed, indicating that additional factors also influence housing values.

Inference:

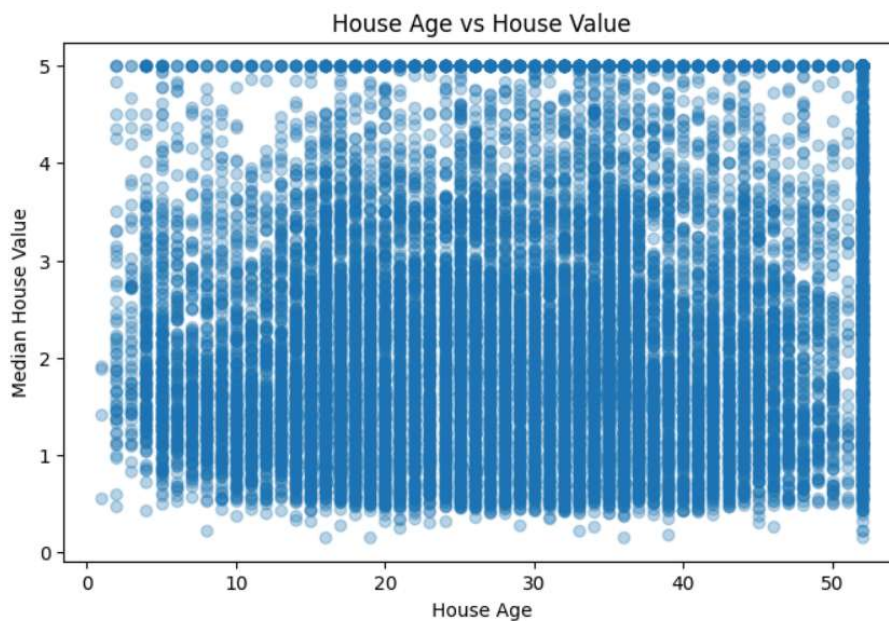
The positive trend confirms the strong relationship identified in the correlation heatmap. Median income appears to be one of the most important predictors of house prices and is expected to contribute significantly to the Linear Regression model.

Scatter Plot: House Age vs House Value

A scatter plot was generated to study the relationship between the age of houses and their market value.

```
plt.scatter(df['HouseAge'], df['MedHouseVal'])
```

The plot helps determine whether older or newer houses tend to have higher house values.



Observations:

- The points were widely scattered across the graph.
- A weak positive relationship was observed between house age and house value.
- House prices varied considerably even for houses of similar age.
- Some older houses exhibited high property values, while some newer houses had comparatively lower values.

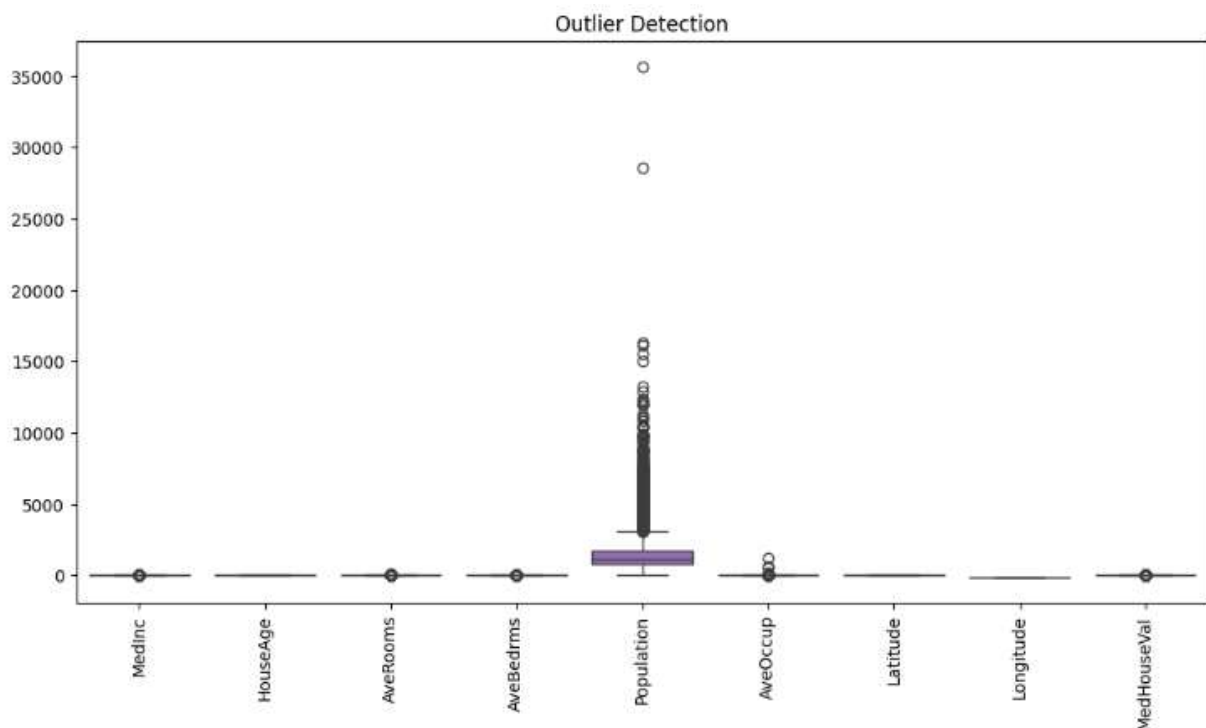
Inference:

The scatter plot indicates that House Age alone is not a strong predictor of house value. Although older houses may sometimes have higher values due to location, infrastructure, or historical significance, house prices are influenced by multiple factors. Compared to Median Income, House Age shows a much weaker relationship with the target variable

Box Plot for Outlier Detection

Box plots were used to identify unusual observations and extreme values in the dataset.

```
sns.boxplot(data=df)
```



Observations:

- Several features contained values outside the normal range.
- Population and occupancy-related variables showed noticeable outliers.

Inference:

Outliers may affect model performance and should be monitored during analysis. However, since Linear Regression can still provide a useful baseline model, the dataset was retained without removing outliers for this assignment.

Overall Findings from EDA

The exploratory analysis provided several important insights:

1. No significant missing values were found in the dataset.
2. Median Income showed the strongest positive relationship with house value.
3. House Age exhibited only a weak relationship with house value.
4. Geographic features (Latitude and Longitude) also influence housing prices.
5. Some variables contained outliers and skewed distributions.
6. The dataset is suitable for applying Linear Regression for house price prediction.

Feature Selection

Feature selection is the process of identifying the input variables that will be used to train the machine learning model. In this project, all housing attributes were selected as input features because they provide useful information for predicting house prices.

```
X = df.drop("MedHouseVal", axis=1)
```

```
y = df["MedHouseVal"]
```

- **X** contains all the independent variables (features).
- **y** contains the target variable, **Median House Value (MedHouseVal)**.
- The target column is removed from the feature set to prevent data leakage.

Train-Test Split

Before training the model, the dataset was divided into training and testing subsets. This helps evaluate the model's ability to make predictions on unseen data.

```
X_train, X_test, y_train, y_test = train_test_split( X,y, test_size=0.2,random_state=42)
```

- `test_size=0.2` allocates 20% of the data for testing and 80% for training.
- `random_state=42` ensures reproducibility by producing the same split every time the code is executed.
- The training dataset is used to learn the relationship between features and house prices.
- The testing dataset is reserved for evaluating model performance.

Model Training

A Linear Regression model from the Scikit-learn library was used.

```
model = LinearRegression()
```

```
model.fit(X_train, y_train)
```

- `LinearRegression()` creates an instance of the Linear Regression model.
- `fit(X_train, y_train)` trains the model using the training dataset.
- During training, the algorithm determines the optimal coefficients and intercept that minimize prediction error.
- These coefficients form the **line of best fit**, representing the relationship between the input features and house prices.

```
#Model training
model=LinearRegression()
model.fit(X_train,y_train)
```

... `LinearRegression` ⓘ ⓘ

```
LinearRegression()
```

`model.coef_`

```
array([ 4.48674910e-01,  9.72425752e-03, -1.23323343e-01,  7.83144907e-01,
        -2.02962058e-06, -3.52631849e-03, -4.19792487e-01, -4.33708065e-01])
```

`model.intercept_`

```
np.float64(-37.02327770606409)
```

Prediction

After training the model, predictions were generated for the testing dataset to evaluate how accurately the model could estimate house prices.

```
y_pred = model.predict(X_test)
```

- `predict()` uses the trained Linear Regression model to estimate house values for the test data.
- The predicted values are stored in `y_pred`.
- These predictions are compared with the actual house values (`y_test`) to evaluate performance.

Model Performance Results

The trained Linear Regression model was evaluated using three standard regression metrics.

1. Mean Absolute Error (MAE)

```
mae = mean_absolute_error(y_test, y_pred)
```

MAE measures the average absolute difference between actual and predicted values

- Lower MAE indicates better prediction accuracy.
- It represents the average prediction error made by the model.

Mean Absolute Error: 0.5332001304956553

2. Root Mean Squared Error (RMSE)

```
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
```

RMSE measures the standard deviation of prediction errors.

- Lower RMSE indicates better model performance.
- RMSE penalizes large prediction errors more heavily than MAE.

Root Mean Squared Error: 0.7455813830127764

3.R² Score

```
r2 = r2_score(y_test, y_pred)
```

R² Score measures how well the model explains the variation in the target variable.

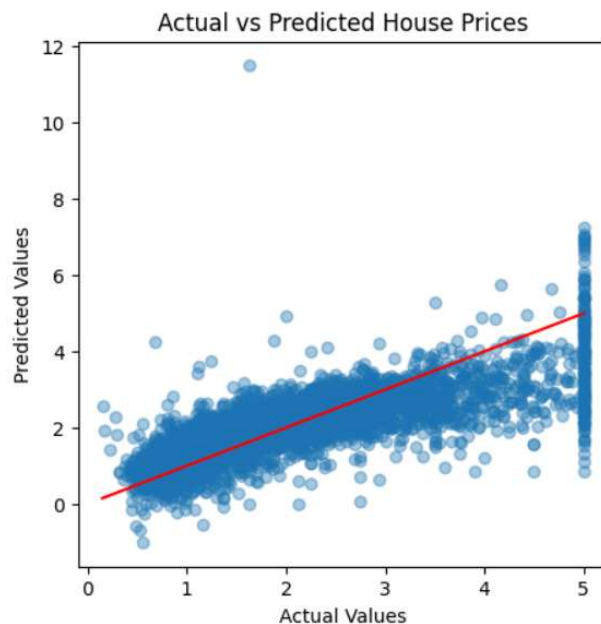
- R² = 1 indicates perfect prediction.
- R² = 0 indicates the model performs no better than using the mean value.
- Higher R² values indicate a better-fitting model.

R-squared: 0.5757877060324508

Actual vs Predicted Plot

The Actual vs Predicted scatter plot was generated to visually compare the model predictions with the true house values.

```
plt.scatter(y_test, y_pred)
```



Observations:

- Many points were clustered around the red reference line.
- Some points deviated from the line, indicating prediction errors.

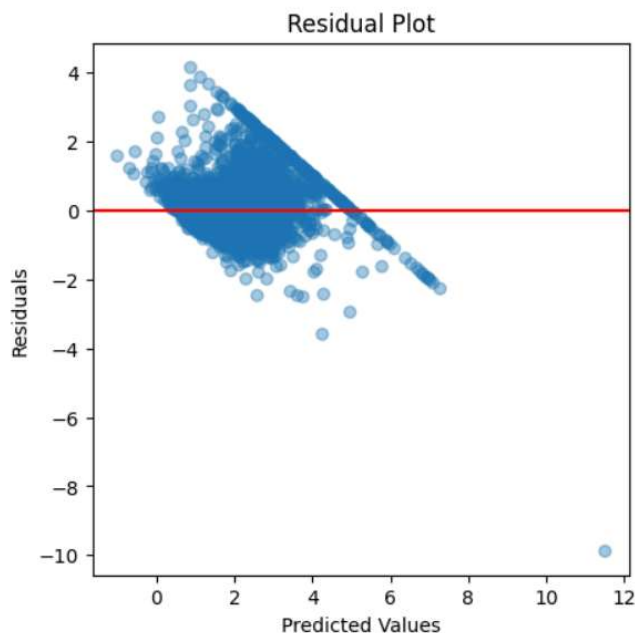
Inference:

The closer the points are to the reference line, the more accurate the predictions. Since most points lie near the line, the model demonstrates reasonable predictive capability and successfully captures the general relationship between housing features and house prices.

Residual Plot

The residual plot was generated to analyze the prediction errors of the model.

```
residuals = y_test - y_pred  
plt.scatter(y_pred, residuals)
```



Observations:

- Residuals were distributed around the zero line.
- No strong systematic pattern was observed.
- Most errors were relatively small.

Inference:

A random distribution of residuals around zero indicates that the Linear Regression model is appropriate for the dataset. The absence of clear patterns suggests that the model captures the major trends in the data and does not suffer from significant bias.

Prediction on New Input Data

To demonstrate the practical use of the trained model, a new set of housing feature values was provided as input to the model.

```
MedInc = float(input("Median Income: "))
HouseAge = float(input("House Age: "))
AveRooms = float(input("Average Rooms: "))
AveBedrms = float(input("Average Bedrooms: "))
Population = float(input("Population: "))
AveOccup = float(input("Average Occupancy: "))
Latitude = float(input("Latitude: "))
Longitude = float(input("Longitude: "))

new_house = [[MedInc, HouseAge, AveRooms, AveBedrms,
               Population, AveOccup, Latitude, Longitude]]

prediction = model.predict(new_house)

print("Predicted House Value:", prediction[0])
```



```
Median Income: 8.3252
House Age: 41
Average Rooms: 6.98
Average Bedrooms: 1.02
Population: 322
Average Occupancy: 2.55
Latitude: 37.88
Longitude: -122.23
Predicted House Value: 4.149487820720765
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739:
  warnings.warn(
```

Conclusion

The Linear Regression model was successfully trained using the California Housing Dataset. Evaluation metrics and visualization techniques confirmed that the model achieved satisfactory predictive performance. The Actual vs Predicted plot demonstrated reasonable accuracy, while the Residual Plot indicated that prediction errors were randomly distributed, validating the suitability of Linear Regression for this house price prediction task.